Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2011

BIOLOGY — **9648/02**

15 AUGUST 2011

HIGHER 2 — **2 hours**

Paper 2 Core Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST
Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
8 / 9	
Total	**100**

This question paper consists of **19** printed pages.

1 (a) Fig. 1.1 below is an electron micrograph showing part of an animal cell.

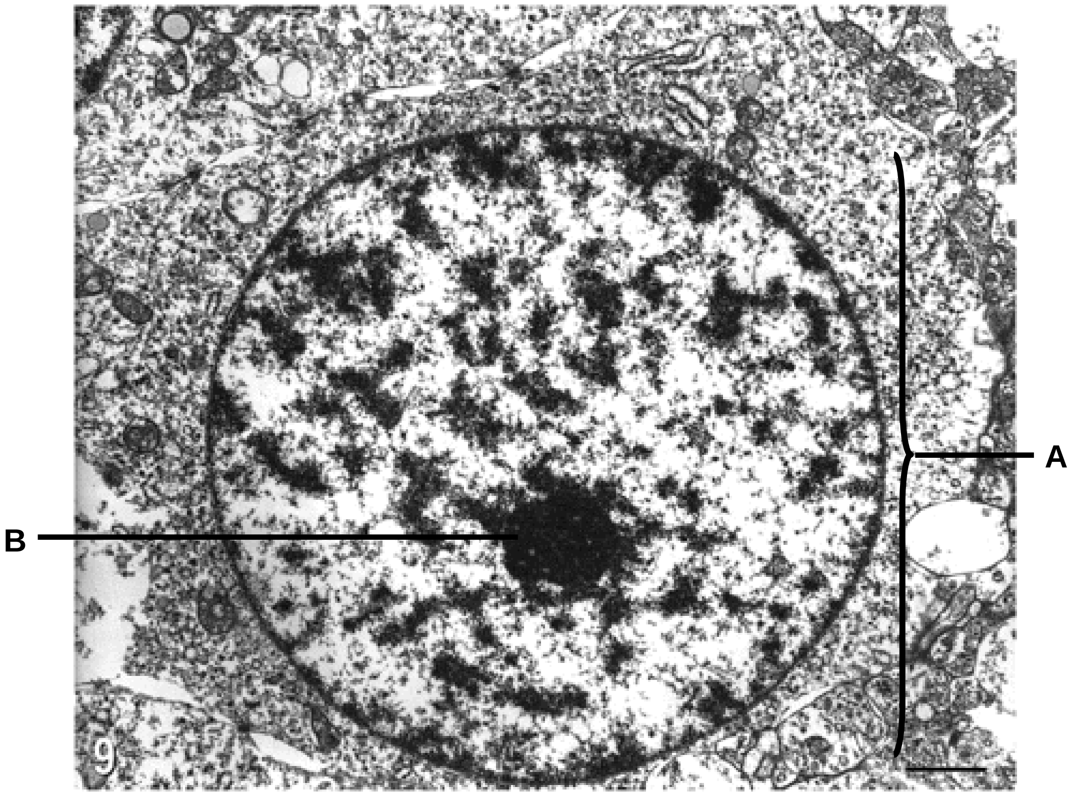


Fig. 1.1

(i) Name the structure labelled **B**.

B: .... [1]

(ii) State **one** visible feature that enabled your identification of structure **B** and describe the role of structure **B**.

....

....

.... [2]

(iii) Describe the roles of mRNAs and rRNAs in protein synthesis.

....

....

....

....

....

.... [4]

For examiner's use

(b) During cell division, structures **D** and **E** shown in Fig. 1.2 are prominent in organelle **A**.

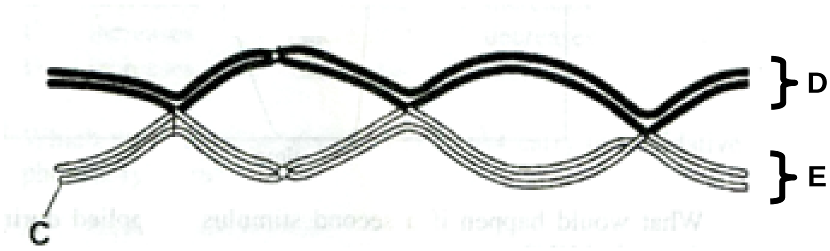


Fig. 1.2

(i) Identify the stage of cell division shown in Fig. 1.2.

.... [1]

(ii) Describe **one** way in which structures **D** and **E** shown in Fig. 1.2 are genetically similar to each other and **one** way in which they are genetically different from each other.

Similarity ....

....

Difference ....

.... [2]

(iii) Structure **C** is made up of protein and non-protein materials. Describe the structure of the non-protein material.

....

....

....

....

.... [3]

[Total: 13]

2 Fig. 2.1 shows the number of HIV RNA copies and $CD4^+$ T cell counts over the course of a typical HIV infection.

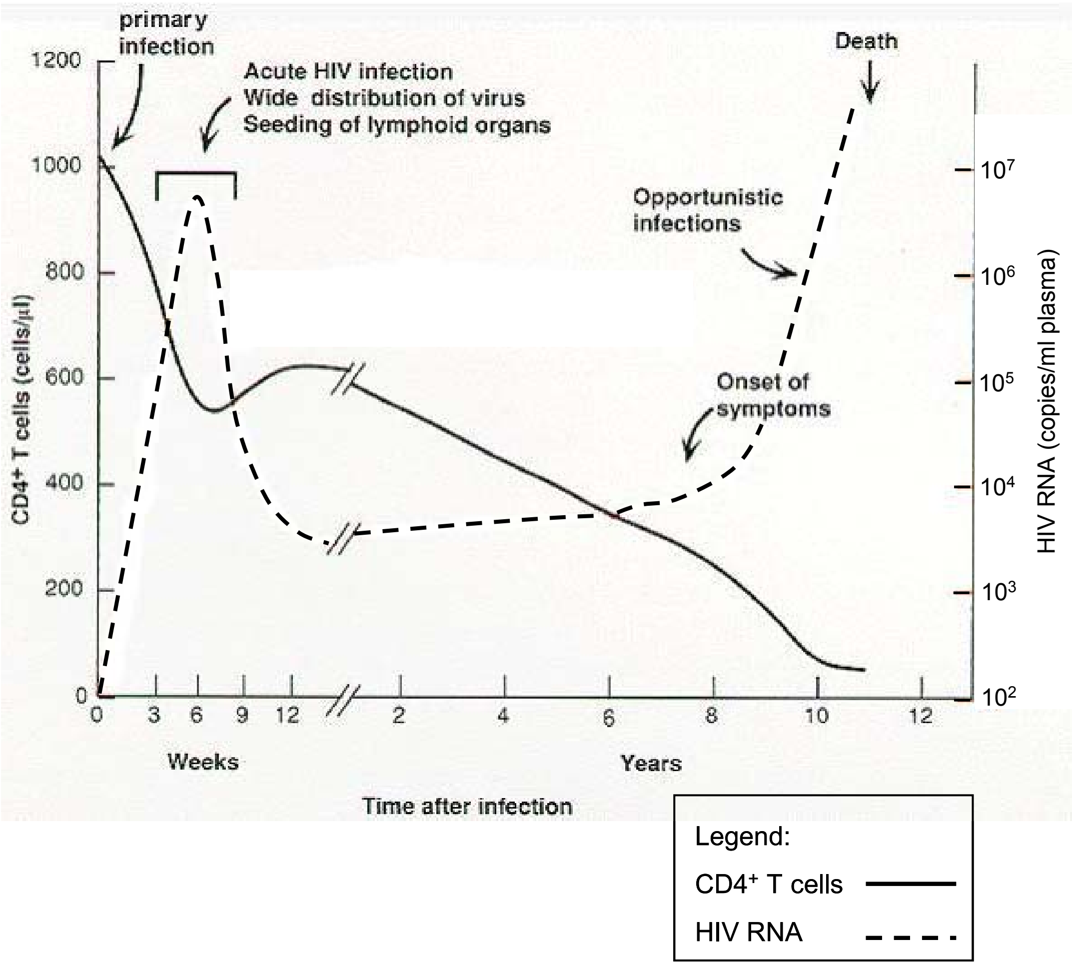


Fig. 2.1

(a) Describe how the HIV genome enters a $CD4^+$ T cell.

....

....

.... [2]

(b) With reference to Fig. 2.1,

(i) describe and explain the changes in $CD4^+$ T cell counts and HIV RNA copies during the first six weeks of infection.

....

....

.... [2]

For examiner's use

(ii) suggest an explanation for the changes in $CD4^+$ T cell counts and HIV RNA copies between week 6 to the point where the graph breaks.

....

....

.... [2]

Integrase is one of the key targets for anti-retroviral therapy. An example of an anti-retroviral drug that acts on integrase is raltegravir. Fig. 2.2 shows the normal reaction catalysed by integrase, and Fig. 2.3 shows the effect of raltegravir on integrase.

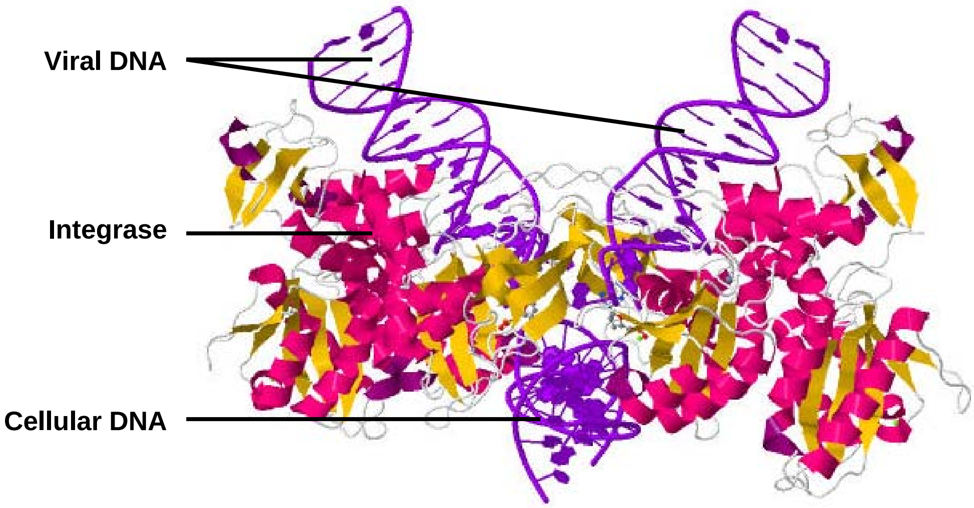


Fig. 2.2

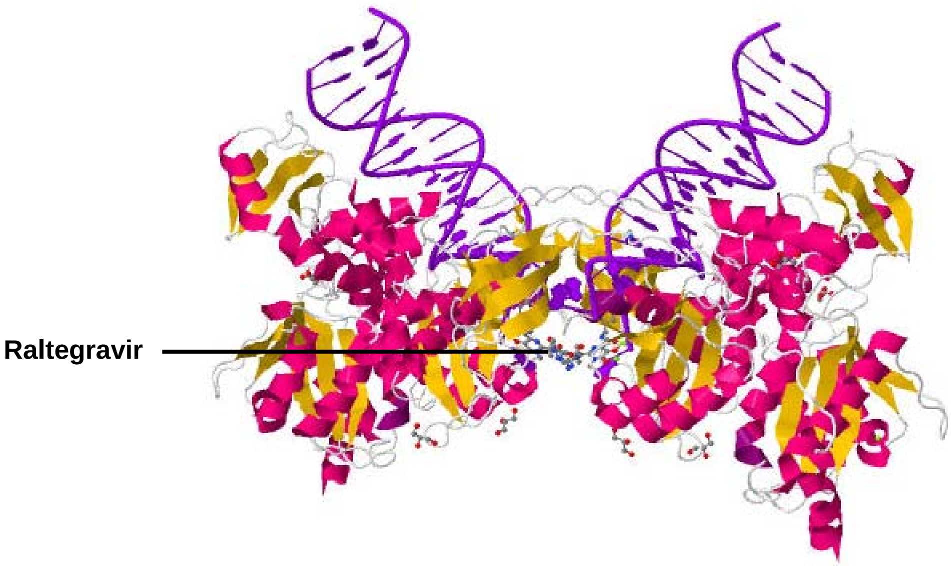


Fig.2.3

For examiner's use

(c) Based on Fig. 2.2 and Fig. 2.3, describe the mode of action of raltegravir on integrase.

....

....

....[2]

(d) Suggest how raltegravir prevents the spread of HIV to other T cells during the acute HIV infection period (weeks 3 – 9 in Fig. 2.1).

....

....

....[2]

[Total: 10]

For examiner's use

3 Fig. 3.1 outlines the production of a protein in a eukaryotic cell.

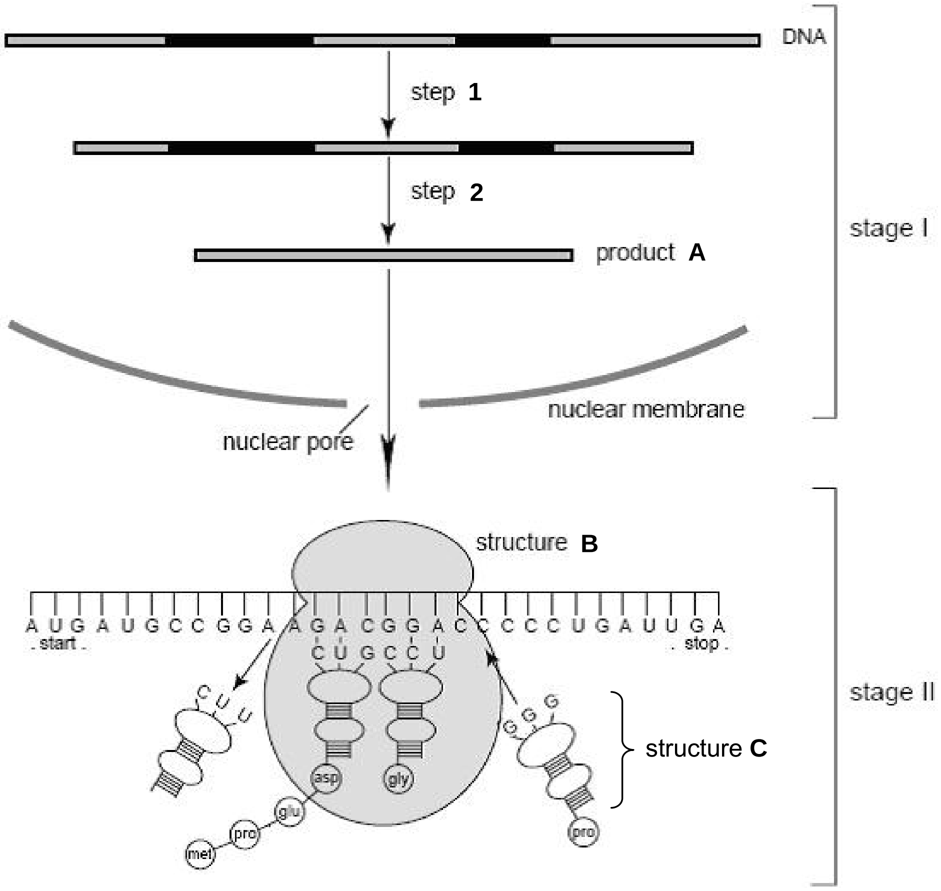


Fig. 3.1

(a) Identify structures **A** and **B**.

A ______________________ **B** ______________________ [1]

(b) Explain how the structure of **C** is adapted to its function.

....

....

.... [2]

(c) Starting from the position of the ribosome as shown in stage II in Fig. 3.1, outline the steps that occur to produce the complete polypeptide.

....[3]

The unfolded protein response (UPR) is a cellular stress response that is triggered by an accumulation of unfolded or misfolded proteins. One of the primary aims of the UPR is to stop protein translation. During the UPR, a kinase known as PERK is activated and acts on the protein eIF2. The normal function of eIF2 is to bind to a tRNA with the UAC anticodon and bring it to the ribosome.

(d) (i) Using the information provided, suggest the role of eIF2 in translation.

....[2]

(ii) Hence suggest how activation of the UPR stops translation.

....[2]

(iii) Apart from the mechanism in **(d)(ii)**, describe **one** other method by which translation can be stopped.

....[1]

(e) Misfolded proteins are eventually degraded in the cytoplasm. Describe how this degradation occurs.

....[1]

[Total: 12]

For examiner's use

4 **(a)** Tail length in cats is determined by the Japanese Bobtail gene. The normal wild-type allele **T** is dominant and codes for normal tail length in cats. The mutant allele **t** is recessive and codes for a short tail, as found in the Japanese Bobtail breed, hence the name.

Coat colour in cats is determined by the *tyr* gene, which codes for tyrosinase enzyme. This enzyme is involved in pigment production, and controls the intensity of body colour in cats. There are five different alleles for the *tyr* gene, namely full colour **B**, Burmese $\mathbf{b^M}$, Siamese $\mathbf{b^S}$, blue-eyed albino $\mathbf{b^A}$, and albino **b**. Each allele gives rise to a different mutant tyrosinase enzyme, resulting in the different phenotypes observed.

Table 4.1 below shows the different breeds possible.

Table 4.1

	B	$\mathbf{b^M}$	$\mathbf{b^S}$	$\mathbf{b^A}$	**b**
B	Normal	Normal	Normal	Normal	Normal
$\mathbf{b^M}$		Burmese	Tonkinese	Burmese	Burmese
$\mathbf{b^S}$			Siamese	Siamese	Siamese
$\mathbf{b^A}$				Blue-Eyed Albino	Blue-Eyed Albino
b					Albino

(i) With reference to Table 4.1 above, determine the relative dominance of the alleles **B**, $\mathbf{b^M}$ and $\mathbf{b^S}$.

....

.... [1]

For examiner's use

(ii) A homozygous Burmese female cat with normal tail was crossed with a homozygous Siamese male cat with a short tail. The F_1 offspring were then sibling-mated to form the F_2 generation. Draw a genetic diagram to show the expected genotypes and phenotypes of the F_2 offspring.

[4]

Siamese cats have a unique coat pattern. They have a white body and black extremities. The extremities, which include ears, face, legs and tail, are cooler regions. These cats possess the *tyr* allele $\mathbf{b^s}$**,** which codes for a temperature-sensitive mutant tyrosinase enzyme. Fig. 4.2 below shows a diagram of a typical Siamese cat.

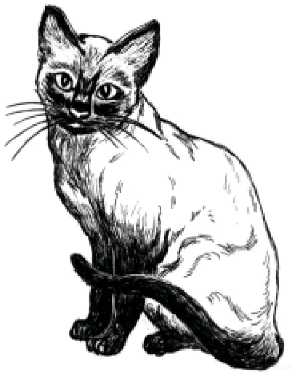


Fig. 4.2

(iii) Suggest how the environment results in the characteristic coat pattern in Siamese cats.

....

....

.... [2]

(b) Tortoiseshell cats are usually female, with a combination of black and orange fur. During skin cell development in the foetal stage, one of the X chromosomes in every cell will be randomly inactivated. The allele on the remaining X chromosome is expressed. The allele for orange fur ($\mathbf{X^O}$) is codominant to the allele for black fur ($\mathbf{X^B}$).

(i) State the probability of producing a female tortoiseshell kitten, when a female tortoiseshell cat mates with a black male cat.

.... [1]

(ii) Approximately 1 in 3000 tortoiseshell cats are male. Based on your knowledge of meiosis, suggest how male tortoiseshell cats may be produced.

....

....

.... [2]

[Total: 10]

For examiner's use

5 Pangolin, armadillo and anteater used to be placed in the same order *Edentata.* Table 5.1 shows some of the characteristics of the three mammals.

Table 5.1

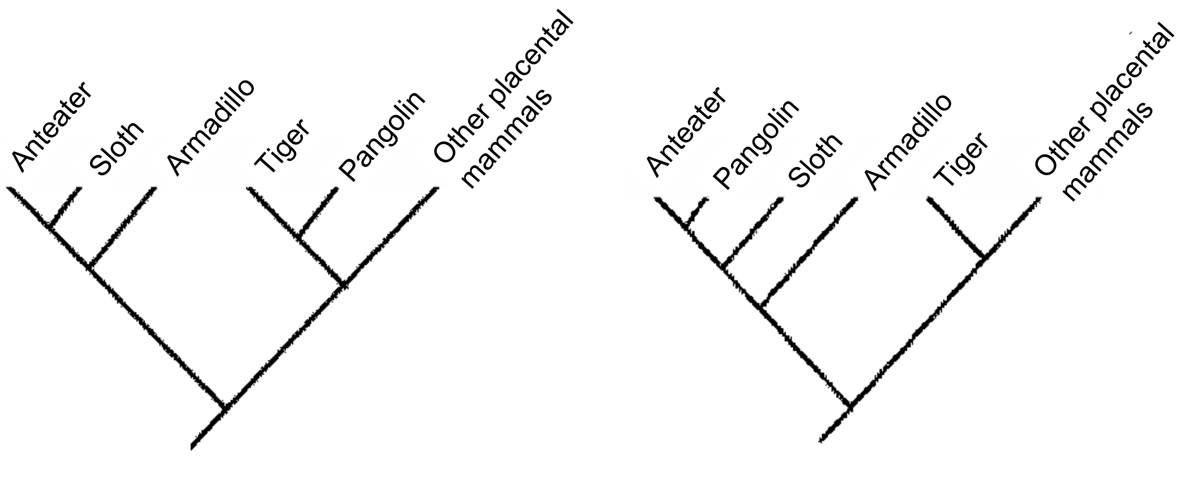
Traits	Pangolin	Armadillo	Anteater
External body	Armoured (with large, hardened, plate-like scales made up of keratin)	Armoured (with carapace comprising of double layer of keratin and bone)	Extensive fur on the body
Diet	Feed only on burrowing social insects such as ants and termites	Feed on insects	Feed on burrowing social insects like ants and termites
Tongue	Long tongues	Long tongues	Long tongues
Limbs	Strong digging limbs	Strong digging limbs	Strong digging limbs

(a) Using information from Table 5.1, explain why pangolin, armadillo and anteater were placed in the same order.

....

....[1]

However, upon the invention of molecular techniques, new light was shed on the classification of pangolin. In the scientists' efforts to reclassify pangolin, armadillo and anteater, two hypothetical phylogenetic trees were constructed as shown in Fig. 5.2.



Phylogenetic tree 1

Phylogenetic tree 2

Fig. 5.2

Fig. 5.3 shows the anatomical structure of the skulls of pangolin, armadillo and anteater, which can be used to support the classification based on molecular techniques.

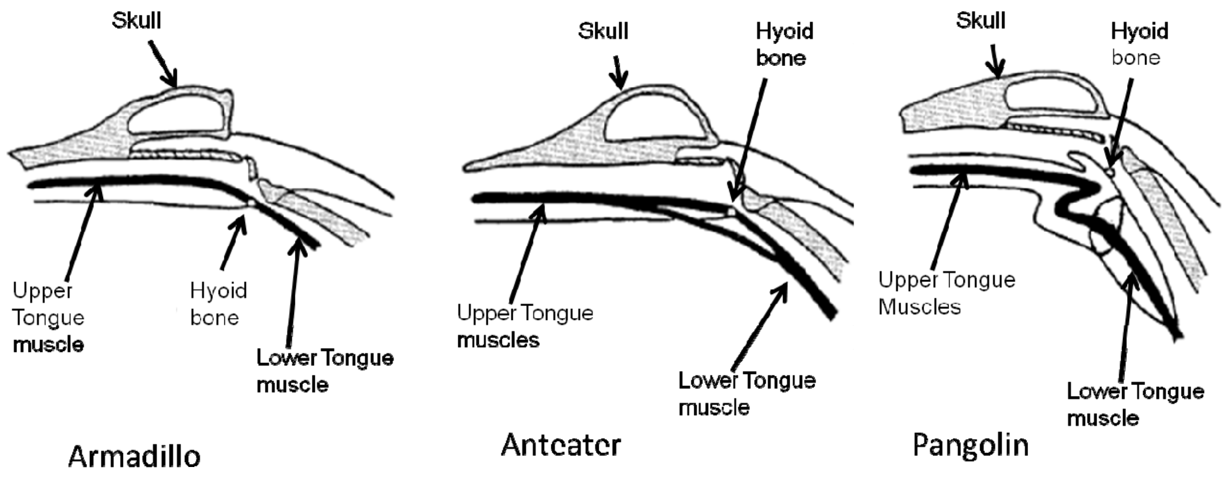


Fig. 5.3

(b) (i) By comparing the structure of the hyoid bone and the tongue muscles between armadillo, anteater and pangolin, state and explain which phylogenetic tree the anatomical structure in Fig. 5.3 supports.

____

____

____

____

____ [3]

(ii) Using your answers in **(a)** and **(b)(i)**, explain how the common features seen in pangolin and armadillo in Table 5.1 came about.

____

____

____ [2]

(c) State **one** other homology that scientists can use to determine the evolutionary relationship between pangolin, armadillo and anteater.

____ [1]

For examiner's use

Table 5.4 shows the timing of the earliest known fossils of pangolin, armadillo, anteater, and the locations in which they were found. The continents of South America and Eurasia (Europe) were completely separated by 110 million years ago (mya).

Table 5.4

Animal	Age of earliest fossil / mya	Fossil location
Armadillo	60	South America
Anteater	55	South America
Pangolin	56	Europe
Sloth	54	South America

(d) (i) Explain why fossils can be used as an evidence to support deductions in evolutionary relationship made using homologies.

....

....

.... [2]

(ii) Using information from Table 5.4, describe how evidence of the fossil records supports your answer in **(b)(i)**.

....

....

....

....

.... [3]

[Total: 12]

For examiner's use

6 **(a)** Among the different biological molecules found in living organisms, both starch and haemoglobin are complex and large biological polymers which play different roles.

Describe **three** structural differences between starch and haemoglobin.

....

....

....

....

.... [3]

(b) Insulin and glucagon are the key hormones for regulating blood glucose levels. They exert their influences on cells by signalling via a membrane-bound receptor.

(i) Name the respective receptors involved in the cell signalling pathway of these hormones.

....

.... [1]

(ii) With reference to cell signalling, state **one** similarity and **two** differences in the mode of action of these hormones.

....

....

....

....

.... [3]

(c) Glucose enters the cells by active transport. Once inside the cell, glucose is broken down to form pyruvate during glycolysis. Fig. 6.1 outlines the processes that occur during glycolysis within the cells.

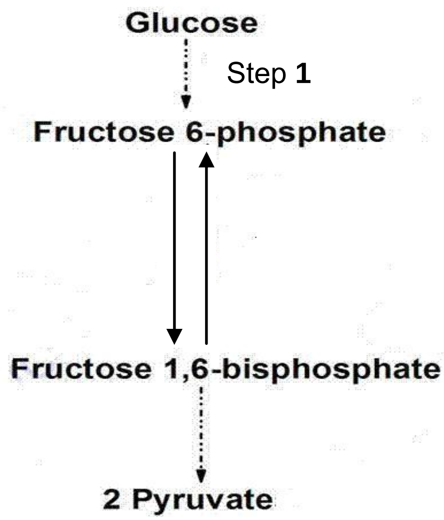


Fig. 6.1

(i) Explain the significance of step **1** in Fig. 6.1.

....

.... [1]

(ii) 2-nitrobenzoic acid is a chemical that binds to NAD^+. Explain the effect of the addition of this chemical on glycolysis.

....

....

....

....

.... [3]

[Total: 11]

For examiner's use

7 Dendrotoxins are a class of neurotoxins found in the venom of snakes belonging to the genus *Dendroaspis*. They bind to voltage-gated potassium channels on myelinated motor neurones. As a result, victims usually suffer from muscle hyperexcitability (a period of excessive excitation to a stimulus), leading to convulsions (uncontrollable shaking of the body).

(a) **(i)** Explain the importance of nodes of Ranvier in the neurone.

[2]

(ii) Explain how a nervous impulse may be generated in a human neurone.

[3]

(iii) Suggest how dendrotoxins cause muscle hyperexcitability. Explain your answer.

[2]

Fig. 7.1 shows a sodium-potassium pump in a section of the plasma membrane belonging to a neurone.

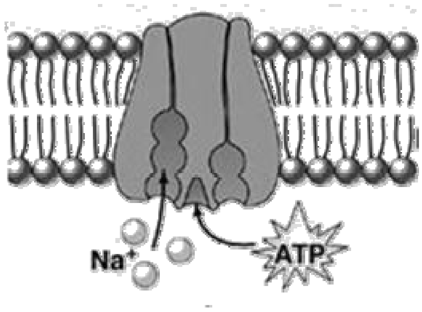


Fig. 7.1

(b) (i) Both ungated ion channels and sodium-potassium pumps are required in maintaining the resting potential of neurones at −70mV. Explain how the sodium-potassium pump achieves its role.

....

....

....

....

.... [3]

(ii) Explain why the plasma membrane allows lipid soluble molecules but not water soluble molecules to pass through.

....

....

.... [2]

[Total: 12]

For examiner's use

Section B

Answer **one** question.

Write your answers in the lined pages provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8 **(a)** Explain how the structure of phospholipids facilitates the process of chemiosmosis in the chloroplast. [7]

(b) Bacteria reproduce asexually, hence they have evolved several mechanisms to transfer genetic material between bacterial cells. Other than transformation and conjugation, describe **two** other ways of genetic transfer in bacteria. [7]

(c) Briefly explain how molecular homology serves as an evidence to support Darwin's theory of descent with modification. [6]

9 **(a)** Explain the importance of producing genetically identical cells in living organisms. [5]

(b) Using the theories of evolution, explain the huge diversity in DNA sequence of haemoglobin across different taxonomic groups. [8]

(c) Describe how a normal cell can form a malignant tumour. [7]